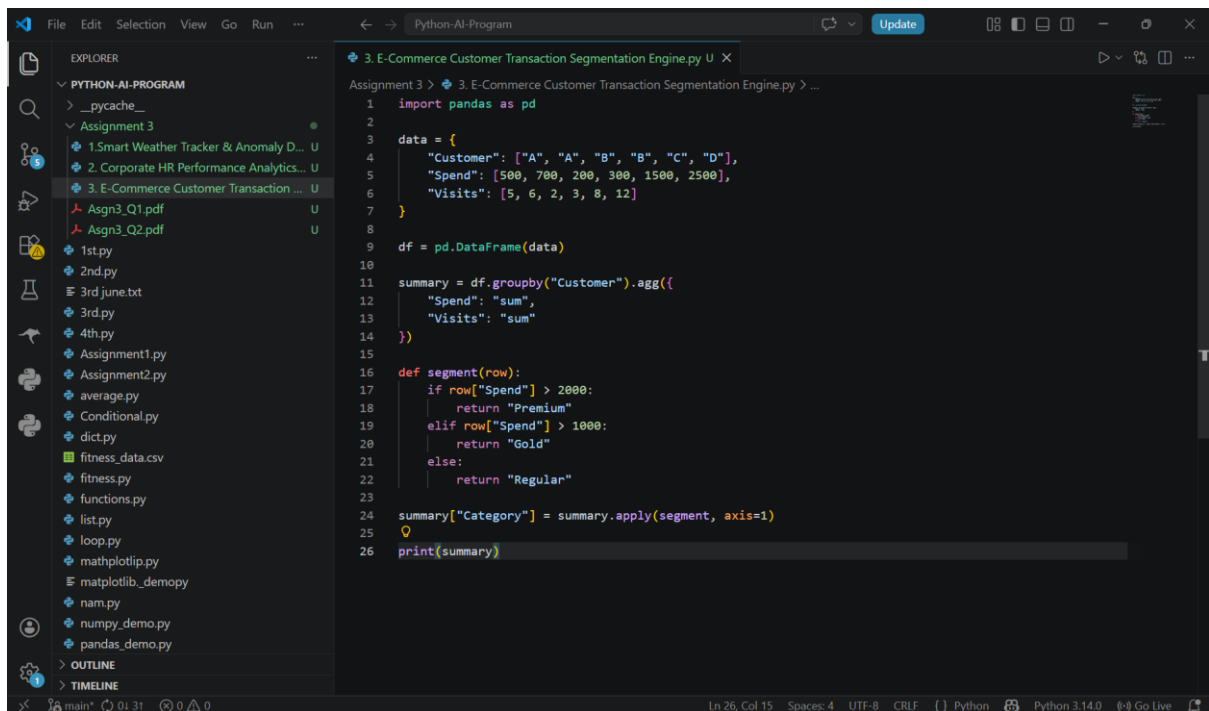
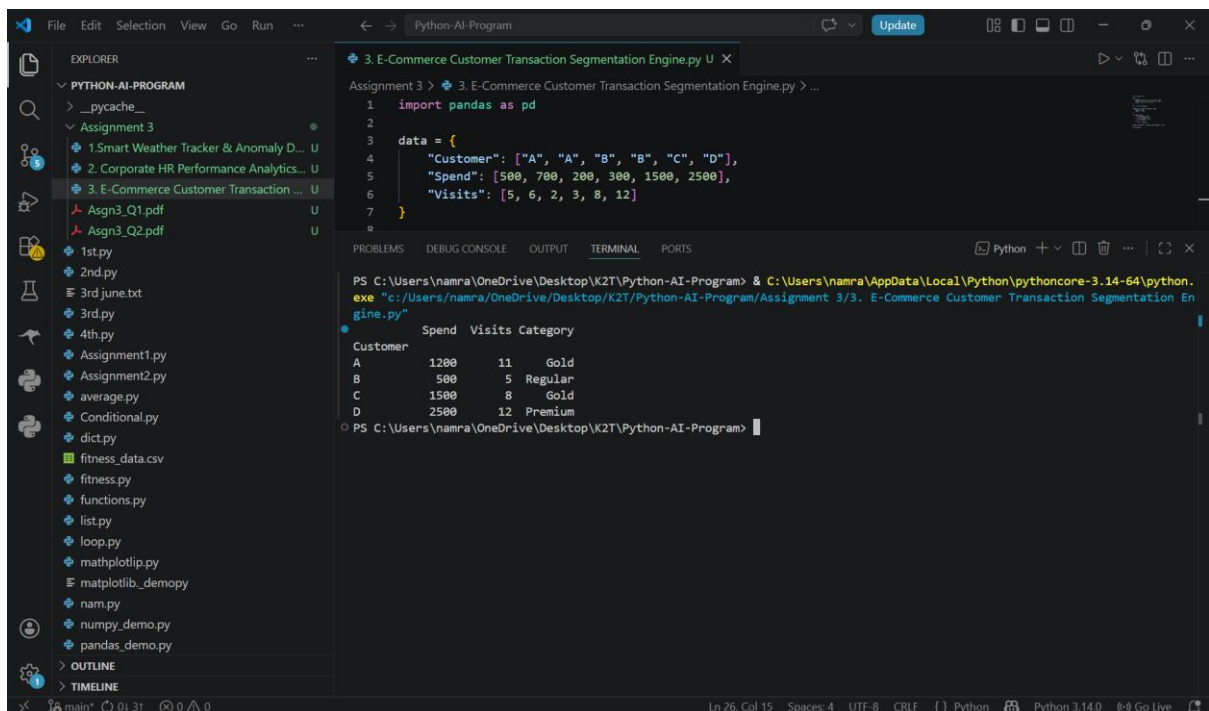


## CODE: 3. E-Commerce Customer Transaction Segmentation Engine



```
1 import pandas as pd
2
3 data = {
4     "Customer": ["A", "A", "B", "B", "C", "D"],
5     "Spend": [500, 700, 200, 300, 1500, 2500],
6     "Visits": [5, 6, 2, 3, 8, 12]
7 }
8
9 df = pd.DataFrame(data)
10
11 summary = df.groupby("Customer").agg({
12     "Spend": "sum",
13     "Visits": "sum"
14 })
15
16 def segment(row):
17     if row["Spend"] > 2000:
18         return "Premium"
19     elif row["Spend"] > 1000:
20         return "Gold"
21     else:
22         return "Regular"
23
24 summary["Category"] = summary.apply(segment, axis=1)
25
26 print(summary)
```

## OUTPUT:



```
PS C:\Users\namra\OneDrive\Desktop\K2T\Python-AI-Program> & C:\Users\namra\AppData\Local\Python\pythoncore-3.14-64\python.exe "c:/Users/namra/OneDrive/Desktop/K2T/Python-AI-Program/Assignment 3/3. E-Commerce Customer Transaction Segmentation Engine.py"

Customer  Spend  Visits  Category
A         1200    11      Gold
B          500     5      Regular
C         1500     8       Gold
D         2500    12      Premium
```